

2016.0 RANGE ROVER (LG), 100-00

GENERAL INFORMATION

DESCRIPTION AND OPERATION

TELEVISION CONTROL MODULE (TVCM) - HYBRID

CAUTION:

Diagnosis by substitution from a donor vehicle is **NOT** acceptable. Substitution of control modules does not guarantee confirmation of a fault, and may also cause additional faults in the vehicle being tested and/or the donor vehicle.

NOTES:

- If a control module or a component is suspect and the vehicle remains under manufacturer warranty, refer to the Warranty Policy and Procedures manual, or determine if any prior approval programme is in operation, prior to the installation of a new module/component.
- Generic scan tools may not read the codes listed, or may read only 5-digit codes. Match the 5 digits from the scan tool to the first 5 digits of the 7-digit code listed to identify the fault (the last 2 digits give extra information read by the manufacturer-approved diagnostic system).
- When performing voltage or resistance tests, always use a digital multimeter accurate to three decimal places, and with an up-to-date calibration certificate. When testing resistance always take the resistance of the digital multimeter leads into account.
- Check and rectify basic faults before beginning diagnostic routines involving pinpoint tests.
- Inspect connectors for signs of water ingress, and pins for damage and/or corrosion.
- If DTCs are recorded and, after performing the pinpoint tests, a fault is not present, an intermittent concern may be the cause. Always check for loose connections and corroded terminals.
- Check DDW for open campaigns. Refer to the corresponding bulletins and SSMs which may be valid for the specific customer complaint and carry out the recommendations as required.

The table below lists all Diagnostic Trouble Codes (DTCs) that could be logged in the Television Control Module (TVCM) - Hybrid. For additional diagnosis and testing information, refer to the relevant Diagnosis and Testing section in the workshop manual.

For additional information, refer to: [Information and Entertainment System](#)

(415-00 Information and Entertainment System - General Information, Diagnosis and Testing).

DTC	DESCRIPTION	POSSIBLE CAUSES	ACTION
B1A56-11	Antenna - Circuit short to ground	NOTE: Circuit reference - ANTENNA_1_SIG Antenna 1 signal circuit short circuit to ground	Refer to the electrical circuit diagrams and check the antenna 1 signal circuit for short circuit to ground
B1A56-12	Antenna - Circuit short to battery	NOTE: Circuit reference - ANTENNA_1_SIG Antenna 1 signal circuit short circuit to power	Refer to the electrical circuit diagrams and check the antenna 1 signal circuit for short circuit to power
B1A56-13	Antenna - Circuit open	NOTE: Circuit reference - ANTENNA_1_SIG Antenna 1 signal circuit open circuit, high resistance	Refer to the electrical circuit diagrams and check the antenna 1 signal circuit for open circuit, high resistance
B1D55-11	Antenna #2 - Circuit short to ground	NOTE: Circuit reference - ANTENNA_2_SIG Antenna 2 signal circuit short circuit to	Refer to the electrical circuit diagrams and check the antenna 2 signal circuit for short circuit to ground

		ground	
B1D55-12	Antenna #2 - Circuit short to battery	NOTE: Circuit reference - ANTENNA_2_SIG Antenna 2 signal circuit short circuit to power	Refer to the electrical circuit diagrams and check the antenna 2 signal circuit for short circuit to power
B1D55-13	Antenna #2 - Circuit open	NOTE: Circuit reference - ANTENNA_2_SIG Antenna 2 signal circuit open circuit, high resistance	Refer to the electrical circuit diagrams and check the antenna 2 signal circuit for open circuit, high resistance
B1D56-11	Antenna #3 Circuit - Circuit short to ground	NOTE: Circuit reference - ANTENNA_3_SIG Antenna 3 signal circuit short circuit to ground	Refer to the electrical circuit diagrams and check the antenna 3 signal circuit for short circuit to ground
B1D56-12	Antenna #3 Circuit - Circuit short to battery	NOTE: Circuit reference - ANTENNA_3_SIG Antenna 3 signal circuit short circuit to power	Refer to the electrical circuit diagrams and check the antenna 3 signal circuit for short circuit to power

B1D56-13	Antenna #3 Circuit - Circuit open	NOTE: Circuit reference - ANTENNA_3_SIG Antenna 3 signal circuit open circuit, high resistance	Refer to the electrical circuit diagrams and check the antenna 3 signal circuit for open circuit, high resistance
B1D57-11	Antenna #4 Circuit - Circuit short to ground	NOTE: Circuit reference - ANTENNA_4_SIG Antenna 4 signal circuit short circuit to ground	Refer to the electrical circuit diagrams and check the antenna 4 signal circuit for short circuit to ground
B1D57-12	Antenna #4 Circuit - Circuit short to battery	NOTE: Circuit reference - ANTENNA_4_SIG Antenna 4 signal circuit short circuit to power	Refer to the electrical circuit diagrams and check the antenna 4 signal circuit for short circuit to power
B1D57-13	Antenna #4 Circuit - Circuit open	NOTE: Circuit reference - ANTENNA_4_SIG Antenna 4 signal circuit open circuit, high resistance	Refer to the electrical circuit diagrams and check the antenna 4 signal circuit for open circuit, high resistance
B1D58-11	Television Output - Circuit short to		Refer to the electrical circuit diagrams and check the CVBS signal circuit for short circuit to

	ground	NOTE: Circuit reference - CVBS OUT / CVBS GND CVBS signal circuit short circuit to ground	ground
B1D58-12	Television Output - Circuit short to battery	CVBS signal circuit short circuit to power	Refer to the electrical circuit diagrams and check the CVBS circuit for short circuit to power
B1D58-13	Television Output - Circuit open	CVBS signal circuit open circuit, high resistance	Refer to the electrical circuit diagrams and check the CVBS circuit for open circuit, high resistance
U3000-49	Control Module - Internal electronic failure	TV control module failure	- Using the manufacturer approved diagnostic system check and install the latest relevant level of software to the TV control module - Clear the DTCs and retest. If fault persists, check and install new TV control module as required
U3000-4B	Control Module - Over temperature	TV control module over temperature	Check airflow to the TV control module. Using the manufacturer approved diagnostic system, clear the DTCs and retest. If the fault persists, install a new TV control module
U3000-51	Control Module - Not programmed	TV control module is not configured correctly	Using the manufacturer approved diagnostic system, re-configure the TV control module with the latest level software
U3000-	Control	TV control module is	Using the manufacturer

54	Module - Missing calibration	not configured correctly	approved diagnostic system, re-configure the TV control module with the latest level software
U3003- 62	Battery Voltage - Signal compare failure	Mismatch between the voltage at the TV control module and the voltage value broadcast on the CAN bus	Refer to the electrical circuit diagrams and check the TV control module power and ground circuits for open circuit, high resistance